

UML CLASS DIAGRAM-TO-TEXT TRANSLATOR

USER GUIDE

A practical guide to creating, loading and translating UML class diagrams

CREATE → EXPORT SVG → LOAD → TRANSLATE → REVIEW

Prepared for the final UML Class Diagram-to-Text Translator artefact

1. Introduction

The UML Class Diagram-to-Text Translator converts UML class diagrams into textual descriptions. It was developed to support computing instructors in preparing accessible representations of UML class diagrams for visually impaired students.

The artefact processes UML class diagrams exported from UMLet in Scalable Vector Graphics (SVG) format and produces two complementary outputs: a literal translation and a model translation. The literal translation describes the visible structure and arrangement of the diagram, while the model translation describes the UML concepts and relationships represented in the diagram.

Quick workflow: Create a UML class diagram in UMLet, export it as SVG, load the SVG file into the translator, generate a literal or model translation, and review the resulting text file.

2. Requirements

- Microsoft Windows computer.
- UML Class Diagram-to-Text Translator installation package (Application Files, setup.exe and UMLClassDiagramTranslator.application).
- UMLet for creating UML class diagrams.
- Java Runtime Environment (JRE) required by UMLet.
- Permission to read and write files on the computer.

Input format: The artefact is designed to process UML class diagrams exported from UMLet in SVG format.

3. Installation

The artefact is distributed as a ClickOnce deployment package. Keep the complete deployment package together when installing the application. The package contains the Application Files folder, setup.exe and UMLClassDiagramTranslator.application.

1. Download UML-Class-Diagram-Translator.zip from the Final-Artefact repository folder.
2. Extract the ZIP file to a folder on the Windows computer.
3. Keep Application Files, setup.exe and UMLClassDiagramTranslator.application together in the extracted folder.
4. Double-click setup.exe to start the installation.
5. Follow the installation prompts. After installation, start the UML Class Diagram Translator from the Windows Start menu.
6. Select CREATE CLASS DIAGRAM and confirm that UMLet opens with its UML palette visible.

Important: Do not move setup.exe away from the Application Files folder or UMLClassDiagramTranslator.application. The ClickOnce deployment files are intended to remain together during installation.

4. Main interface

Start the UML Class Diagram Translator. The main screen provides the workflow options shown below.

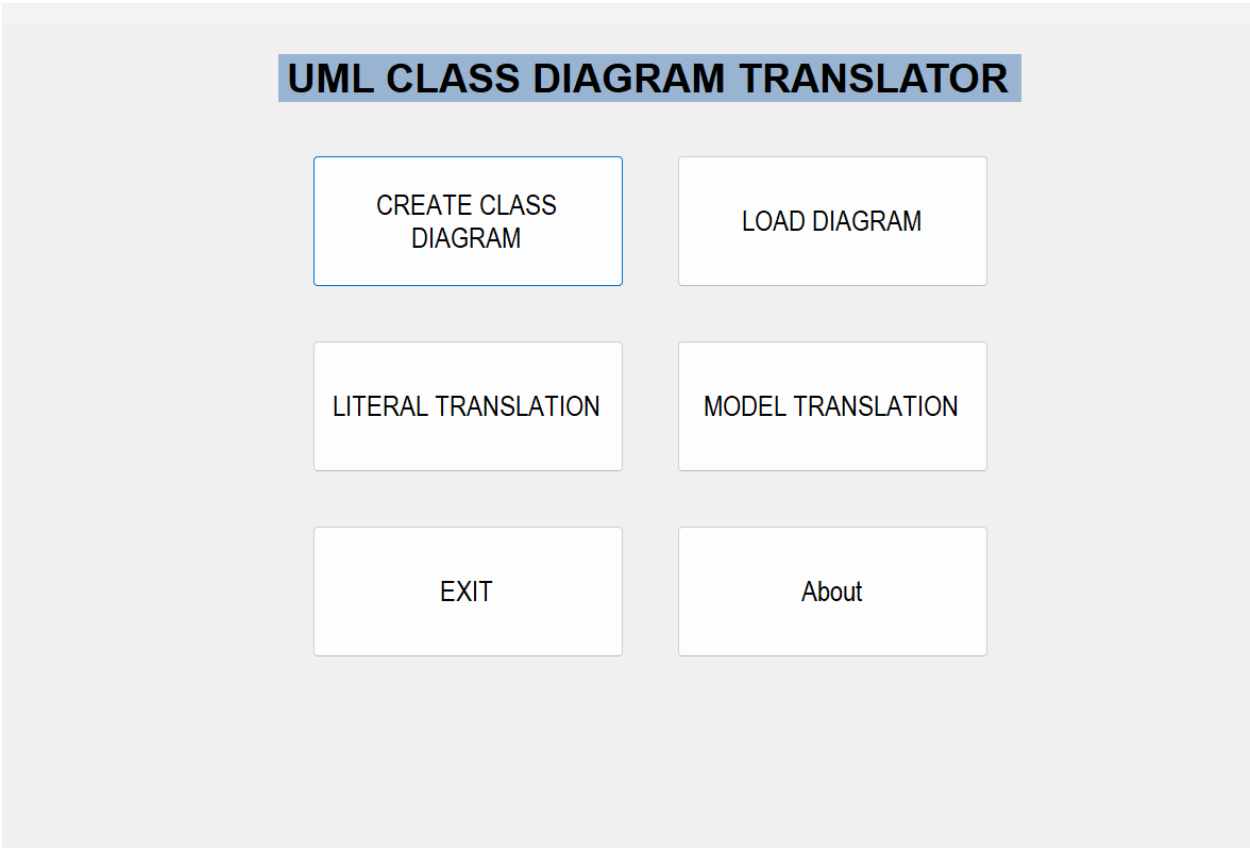


Figure 1: Main interface of the UML Class Diagram Translator

Control	Purpose
CREATE CLASS DIAGRAM	Opens UMLet so that a UML class diagram can be created.
LOAD DIAGRAM	Loads an existing UML class diagram that has been exported as an SVG file.
LITERAL TRANSLATION	Generates a textual description of the graphical structure and layout of the loaded diagram.
MODEL TRANSLATION	Generates a textual description of the UML concepts and relationships represented in the loaded diagram.
EXIT	Closes the application.
About	Displays information about the application.

5. Creating a UML class diagram

A diagram can be created directly from the translator by opening the integrated UMLet environment. The distributed artefact includes the UMLet resources and palettes required by the integrated diagram-creation workflow.

1. Select CREATE CLASS DIAGRAM from the main interface.
2. Create the required UML class diagram in UMLet.
3. Add the required classes, attributes, methods and relationships.
4. Check that the diagram is complete before exporting it.

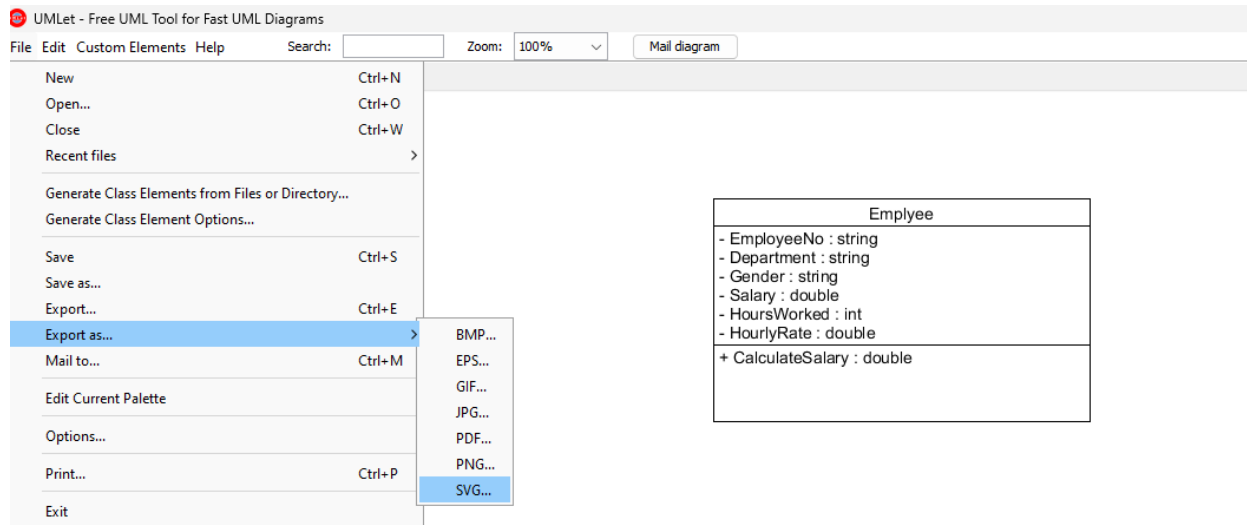


Figure 2: UMLet environment used to create and export a UML class diagram

6. Exporting the diagram as SVG

After the diagram has been completed in UMLet, export it in SVG format and save it in a location that can easily be found. SVG is a structured, text-based graphics format. It retains information about text, shapes, paths and positions that the translator can process.

Important: Use SVG for the diagram that will be loaded into the translator. Other image formats do not provide the structured information used by the artefact.

7. Loading an SVG diagram

1. Return to the UML Class Diagram Translator.
2. Select LOAD DIAGRAM.

3. Browse to the location of the exported SVG file.
4. Select the required SVG file and open it.
5. After the diagram has been processed, choose the required translation option.

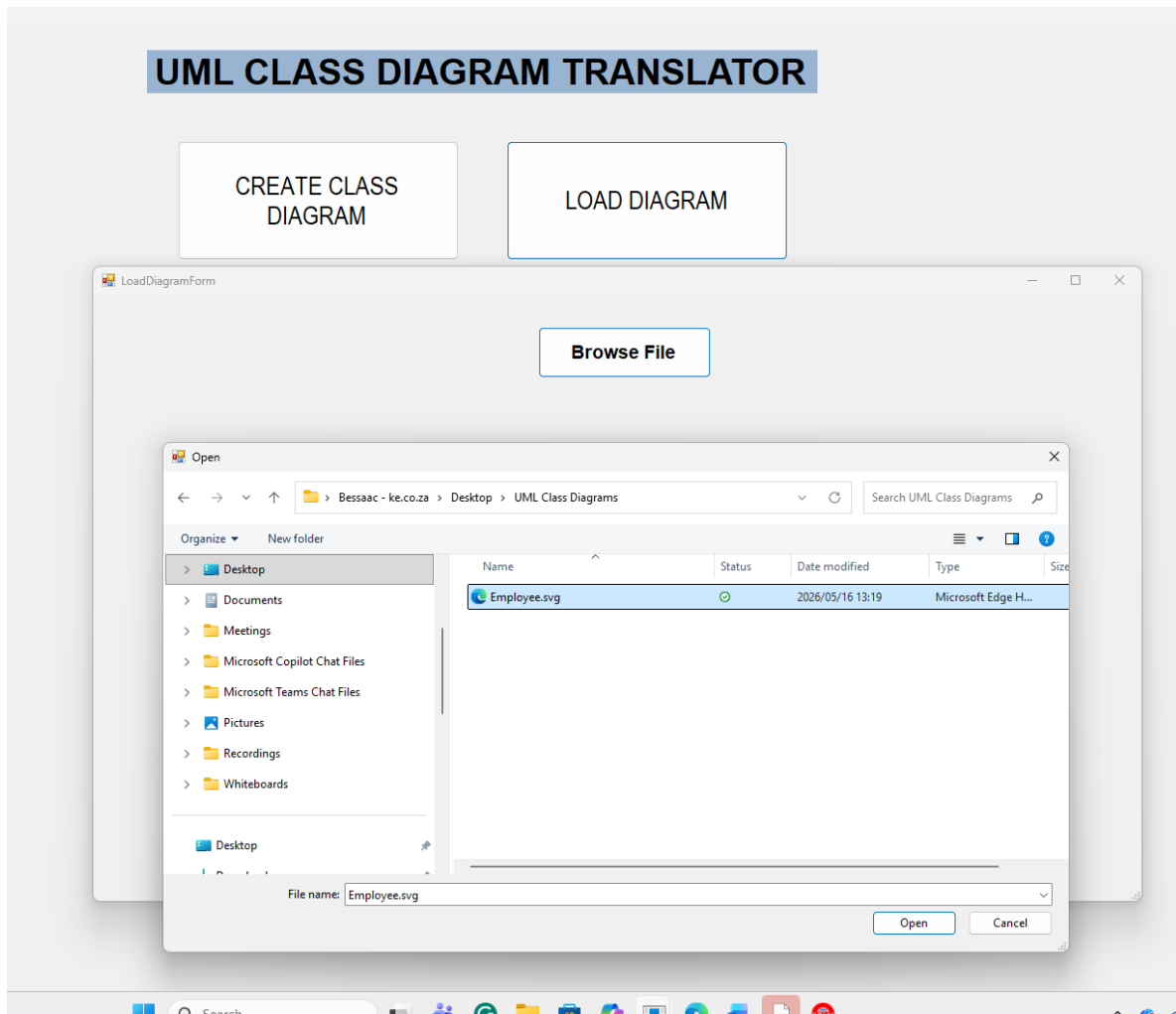


Figure 3: Loading an exported SVG UML class diagram into the translator

8. Generating a literal translation

After loading the SVG file, select LITERAL TRANSLATION. The artefact describes the graphical information represented in the diagram. The literal translation focuses on visible diagram elements and their arrangement rather than on interpreting the UML model semantically.

The output can include text representing class names, attributes, methods, compartments, relationship labels, connectors and other visible elements detected in the SVG file.

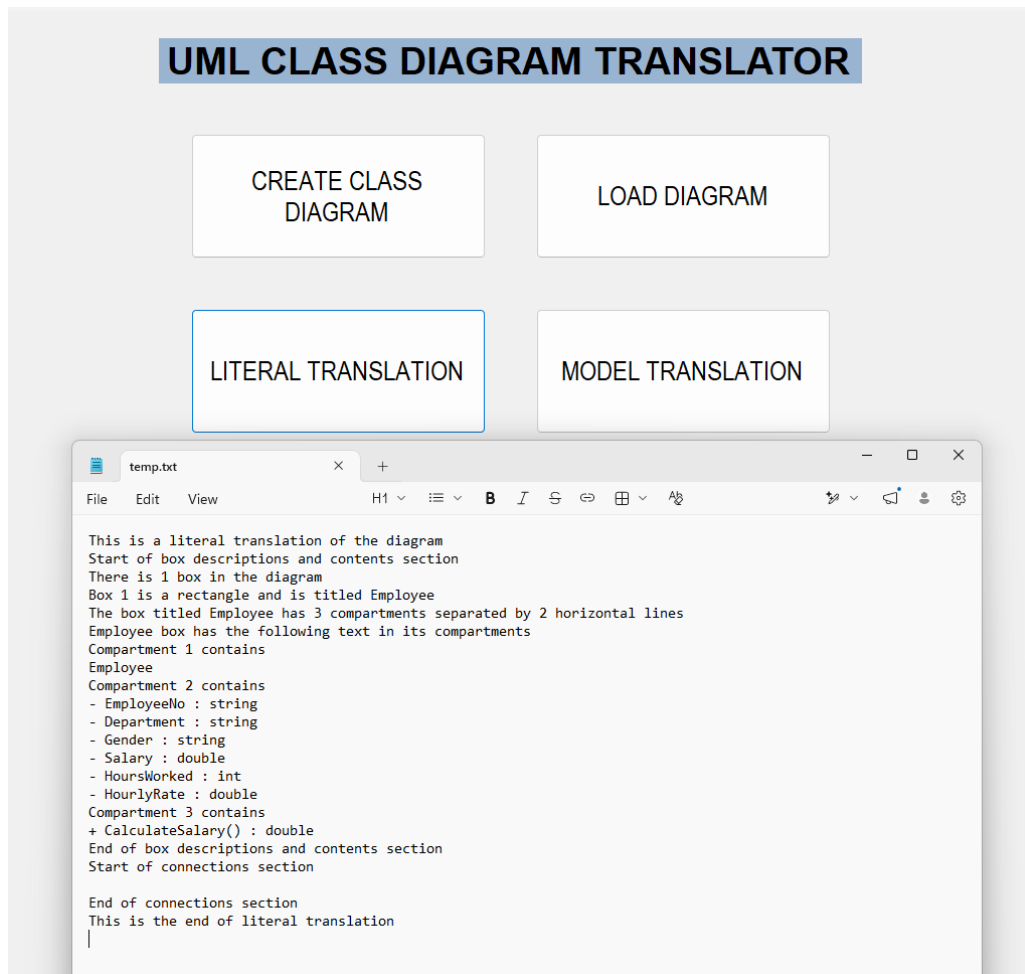


Figure 4: Example literal translation generated from a UML class diagram

9. Generating a model translation

Select MODEL TRANSLATION to generate a semantic description of the loaded UML class diagram. This output interprets the UML notation and describes the meaning of the model.

- classes, attributes and methods;
- associations;
- generalisation relationships;
- aggregation and composition relationships;
- multiplicities; and
- supported self-referencing relationships.

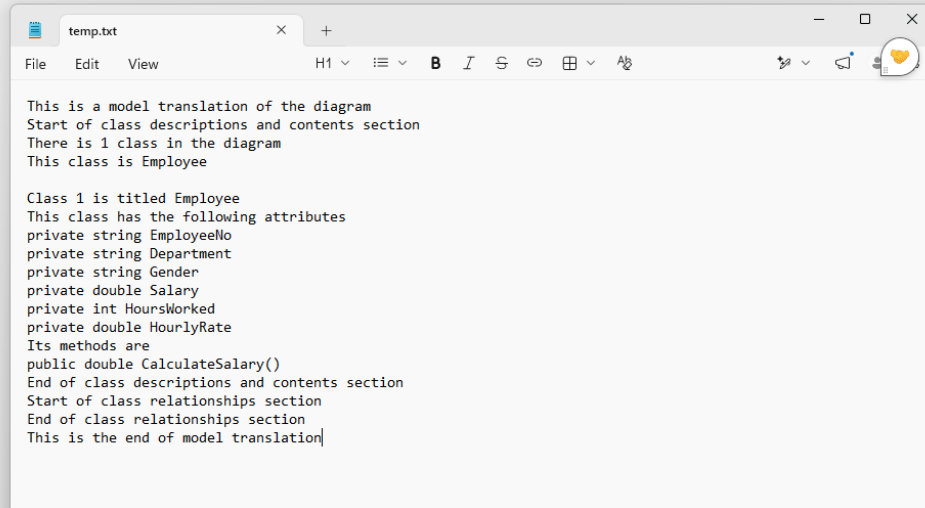
UML CLASS DIAGRAM TRANSLATOR

CREATE CLASS
DIAGRAM

LOAD DIAGRAM

LITERAL TRANSLATION

MODEL TRANSLATION



The screenshot shows a text editor window titled 'temp.txt' with a menu bar (File, Edit, View) and a toolbar. The text content is as follows:

```
This is a model translation of the diagram
Start of class descriptions and contents section
There is 1 class in the diagram
This class is Employee

Class 1 is titled Employee
This class has the following attributes
private string EmployeeNo
private string Department
private string Gender
private double Salary
private int HoursWorked
private double HourlyRate
Its methods are
public double CalculateSalary()
End of class descriptions and contents section
Start of class relationships section
End of class relationships section
This is the end of model translation|
```

Figure 5: Example model translation generated from a UML class diagram

10. How the artefact processes the SVG file

When an SVG file is loaded, the artefact parses its structured content and extracts information required to interpret the diagram. This includes textual content, coordinates and positions, class rectangles and compartments, connector paths, arrowheads, relationship indicators and multiplicity information.

The extracted information is then processed by the translation rules. The literal translation describes the graphical representation, while the model translation interprets the UML meaning represented by those elements.

11. Recommended workflow

Step	Action	Result
1	Create	Create the UML class diagram in UMLet.
2	Export	Export the completed diagram as an SVG file.
3	Load	Load the exported SVG file into the translator.
4	Translate	Generate a literal translation and/or a model translation.
5	Review	Review the generated plain-text output before use or distribution.

12. Output files and accessibility

The generated textual output is stored in plain-text files. Plain text can be opened using standard text-processing software and is suitable for access using screen-reading software. The generated translation should be reviewed before it is distributed as learning material, particularly when the source diagram contains complex UML relationships.

13. Current scope

- UML class diagrams only.
- UMLet SVG files are used as input.
- English textual output.
- Literal and model translation modes.
- Plain-text translation output.

Other UML diagram types, multilingual translation and text-to-UML generation are outside the current scope of the artefact.

14. Closing the application

When the translation task has been completed, select EXIT on the main interface to close the UML Class Diagram Translator.

Good practice: Keep the original UMLet diagram, the exported SVG file and the generated text files together so that the translation can be traced back to its source diagram.